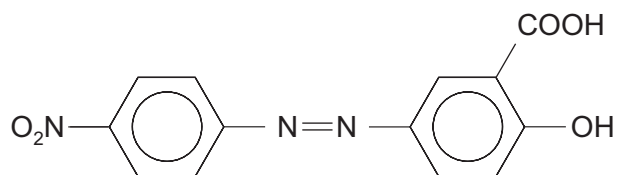


10. (a) The azo dye Mordant Orange 1 has the formula shown below.



It is used as a pH indicator for acid-base titrations. At a pH of 10.1 the dye is yellow but at a pH of 12.1 it is orange-red.

- (i) Give the displayed formulae of the two starting compounds that react, via a diazonium compound, to give Mordant Orange 1. [2]

- (ii) The table shows the colour seen and the colour absorbed for a solution of Mordant Orange 1 as the pH increases.

|                                 |        |               |
|---------------------------------|--------|---------------|
| Colour seen                     | yellow | orange-red    |
| Colour absorbed                 | violet | greenish-blue |
| pH increasing $\longrightarrow$ |        |               |

State how the wavelength of the light **absorbed** changes with pH, giving your reasoning. [1]

.....

.....

.....



- (iii) The visible spectrum of the dye at pH 10.1 shows that it has a maximum absorption at 385 nm.

Calculate the frequency of the radiation being absorbed at 385 nm. [2]

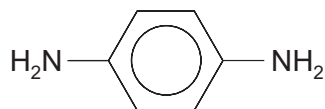
Frequency = ..... Hz

- (iv) Use your answer to (iii) to calculate the energy of the maximum absorption at 385 nm, giving your answer in  $\text{kJ mol}^{-1}$ . [3]

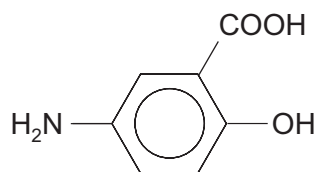
Energy = .....  $\text{kJ mol}^{-1}$



- (b) Mordant Orange 1 can be reduced by the enzymes from certain bacteria to produce 1,4-diaminobenzene and 5-amino-2-hydroxybenzoic acid



1,4-diaminobenzene



5-amino-2-hydroxybenzoic acid

- (i) 1,4-diaminobenzene can also be made from 4-nitrophenylamine.

State the reagent(s) necessary for this reaction.

[1]

- (ii) Use the displayed formula of 1,4-diaminobenzene to help you describe its  $^1\text{H}$  high resolution NMR spectrum. Explain your reasoning.  
*Reference to the position of the signals is not required.*

[3]

- (iii) Give the displayed formula of the organic compound formed when 5-amino-2-hydroxybenzoic acid reacts with aqueous sodium hydroxide in a 1:1 molar ratio.

[1]



- (iv) 5-Amino-2-hydroxybenzoic acid can be used in medicine to treat a number of conditions. It is believed to work by removing free radicals. One of these radicals is the hydroxyl radical. This can be produced by the homolytic fission of hydrogen peroxide,  $\text{H}_2\text{O}_2$ , where the  $\text{O}-\text{O}$  bond is broken.

Draw a dot and cross diagram of the hydroxyl radical, showing the outer electrons for each atom. [1]

Examiner  
only

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